Class 11 Biology Notes

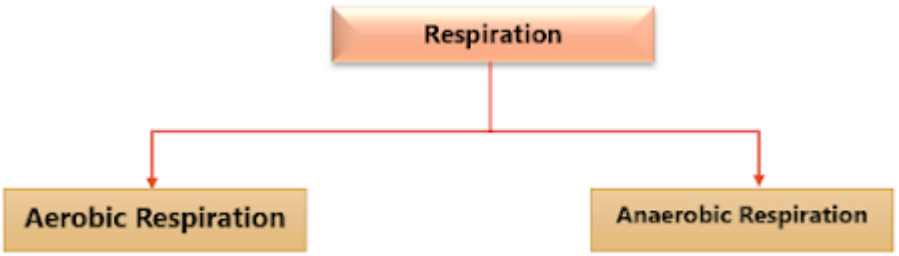
Class 11 Biology Notes Chapter 14 Respiration in Plants contains all the topic as per the syllabus of NCERT. Each topic is explained in very easy language with coloured diagrams. Students can understand Typical topics easily step by step. **Class 11 Biology notes** Chapter 14 Respiration in Plants is prepared by our experts as per the latest syllabus and exam pattern Class 11 Biology. Calculation of problems is given in very simple ways after consulting from various books and expert teachers. Also, you can download Class 11 Notes of other Subjects like Chemistry, Mathematics Formulas sheets, Biology. Along with Notes, we are also providing chapters wise notes of class 11 all other subjects. After reading the Class 11 Biology Notes of Chapter 14 Biology definitely you will get good marks in board exam Class 11 Biology Notes for 14 Respiration in Plants. Each concept is explained in a detailed way class 11 Biology notes for chapter 14 Respiration in Plants notes make according to the Class 11th NCERT Biology book. Students can download pdf file and do study offline so that preparation of exam will be done in better in the bottom section pdf button given so that you have to click on it and can download coloured notes of class 11 Biology notes chapter 14 Respiration in Plants. Notes made according to NCERT Syllabus so that any student who studying in class 11 can read no matter which state you belong to even this notes can help to prepare for completive exam also.if you want to color full notes then **click here**

DOWNLOAD PDF

Class 11 Biology Notes Chapter 14 Respiration in Plants

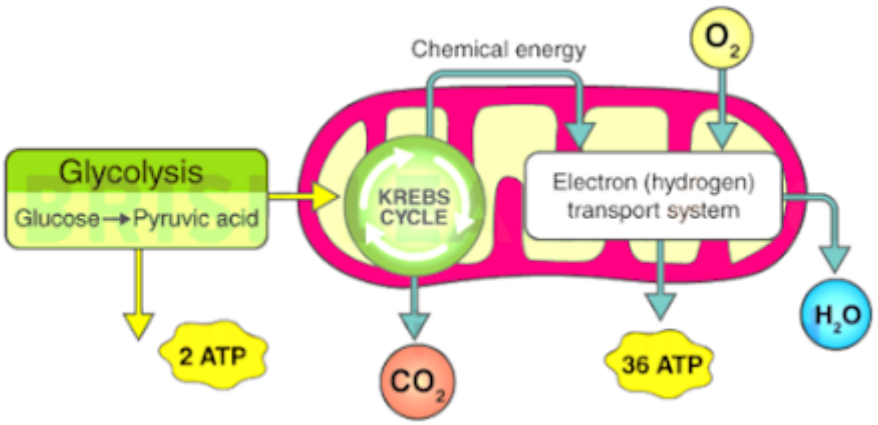
Respiration

Respiration is an energy releasing, enzymatically controlled catabolic process which involves a step-wise oxidative breakdown of food substance inside living cells.



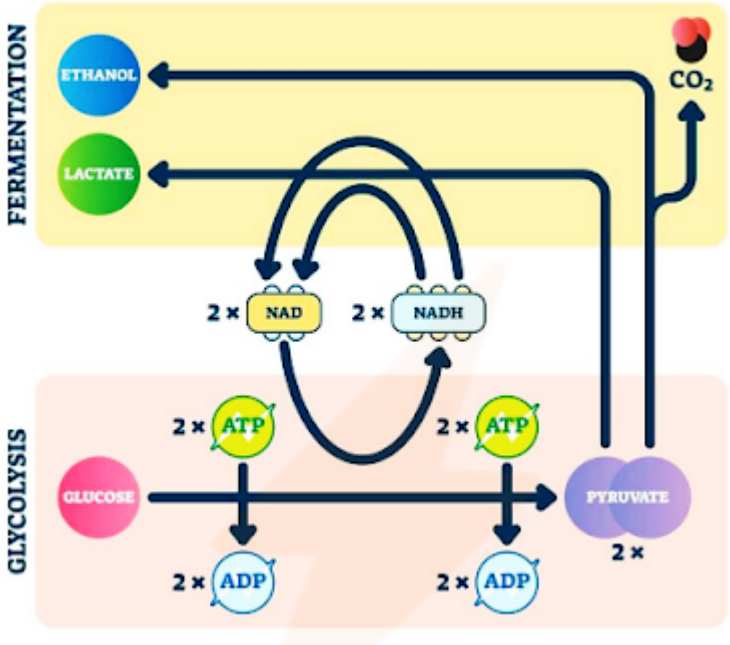
Aerobic Respiration:

Aerobic Respiration is an enzymatically controlled release of energy in a stepwise catabolic process of complete oxidation of organic food into carbon dioxide and water with oxygen acting as terminal oxidant.



Anaerobic Respiration:

Anaerobic respiration is the type of respiration through which cells can break down sugars to generate energy in the absence of oxygen. This is in contrast to the highly efficient process of aerobic respiration, which relies on oxygen to produce energy.



Aerobic Respiration

- Glycolysis
- Kreb's Cycle
- Terminal Oxidation

Glycolysis

- The scheme of glycolysis is given by Gustav Embden, Otto Meyerhof, and J. Parnas. It is also called as EMP pathway.
- Glycolysis is the partial oxidation of glucose or similar hexose sugar into two molecules of pyruvic acid through a series of enzyme mediated reaction releasing some ATP and NADH2. It occurs in cytoplasm.
- In plants glucose is derived from sucrose or from storage carbohydrates. Sucrose is converted into glucose and fructose by enzyme invertase.
- Glycolysis starts with phosphorylation of glucose in presence of enzyme hexokinase to form Glucose-6-phosphate. One molecules of ATP is used in this process.
- In next steps Glucose-6-phosphate is converted into fructose-6-phosphate, catalyzed by enzyme phosphohexose isomerase.
- Fructose-6-phosphate uses another molecule of ATP to form Fructose-1-6 biphosphate in presence of enzyme phosphofructokinase.

Tricarboxylic Acid Cycle/ Kreb's Cycle

- The Acetyl CoA enters a cyclic pathway called TCA cycle or Kreb's cycle.
- TCA cycle was discovered by Hans Krebs in 1940. This cycle is called TCA cycle because initial product is citric acid.
- Acetyl CoA combine with OAA (Oxaloacetic acid) and water to yield citric acid in presence of enzyme citrate synthase to release CoA.
- Citrate is then isomerized to isocitrate. It is followed by two successive steps of decarboxylation, leading to the formation of a-ketoglutaric acid and then succinyl- COA.
- In the remaining steps, succinyl-CoA is oxidized to OAA allowing the cycle to continue.
- There are three points in the cycle where NAD+ is reduced to NADH2 and one point where FAD + is reduced to $FADH_2$.

- A molecule of glucose produces two molecules of $NADH_2$, 2ATP and two pyruvate while undergoing glycolysis. The two molecules of pyruvate are completely degraded in Krebs cycle to form two molecules of ATP, 8NADH, and $2FADH_2$

Terminal Oxidation

Terminal Oxidation is the name of oxidation found in aerobic respiration that occurs towards end of catabolic process and involves the passage of both electrons and protons of reduced coenzyme to oxygen to produce water.

Electron Transport Chain

- The metabolic pathway through which the electron passes from one carrier to another inside the inner mitochondrial membrane is called ETC or mitochondrial respiratory chain.
- Electrons from NADH produced during citric acid cycle are oxidized by NADH dehydrogenase and electrons are transferred to ubiquinone located within the inner membrane. Ubiquinone also receives electrons from FADH2 which is transferred to cytochrome c via cytochrome bc, complex.
- When the electrons pass from one carrier to another via electron transport chain, they produce ATP from ADP and inorganic phosphate. The number of ATP molecules synthesized depends upon electron donor.
- Oxidation of one molecule of NADH gives rise to 3 molecules of ATP, while oxidation of one molecule of FAD_2 produce two molecules of ATP.

Oxidative phosphorylation

It occurs in respiration process. Energy of oxidation-reduction is used for production of proton gradient required for phosphorylation.

Photophosphorylation

It occurs in photosynthesis. Light energy is utilized for production of proton gradient for phosphorylation.

Fermentation

- It accounts for incomplete oxidation of glucose.
- In fermentation, there is net gain of only two molecules of ATP.
- NADH is oxidized to NAD+ very slowly.

Aerobic Respiration

- It accounts for complete oxidation of glucose.
- In aerobic respiration, there is more net gain of ATP.
- NADH is oxidized to NAD+ very fast.

Amphibolic Pathway

- Glucose is the favored substrate for respiration. All carbohydrates are usually converted into glucose before used for respiration.
- Fats needs to be broken down into glycerol and fatty acid, which is further broken converted into Acetyl CoA and enter the respiratory pathway.
- Proteins are broken into amino acids and further enter Krebs cycle.
- Breaking down process within living organism is called catabolism and synthesis process is called anabolism process. So, respiration is an Amphibolic pathway.

PLEASE SHARE THIS

YOU MIGHT ALSO LIKE

Class 11 Biology Notes Chapter – 2 जीव जगत का वर्गीकरण

September 3, 2023

Class 11 Biology Notes Chapter 16 Digestion and Absorption

October 8, 2023

Class 11 Biology Notes Chapter 15 Plant Growth and Development

October 4, 2023